

Testing and Quality Assurance -- Guided Notes

Unit tp-4.3 | CompTIA Network+ N10-009 Obj. FC0-U71 4.3 | N10-008 FC0-U71 4.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

QA (Quality Assurance) is the process of verifying that software meets its _____ before release. A bug (defect) found in _____ is far less expensive to fix than one found in _____. The earlier a defect is found in the SDLC, the _____ it costs to fix.

Question 2

_____ testing verifies each individual function or module works correctly in isolation. _____ testing verifies that components work correctly when combined. _____ testing verifies the complete system meets user requirements from end to end (also called acceptance testing).

Question 3

_____ testing is performed without knowledge of the internal code. The tester interacts with the software as a _____ would. _____ testing is performed with full knowledge of the source code. _____ (gray-box) testing is a combination of both approaches.

Question 4

_____ testing verifies the same software functions correctly across different browsers, operating systems, and hardware. _____ testing subjects the system to heavy load to find failure points. _____ testing verifies performance under expected peak conditions to ensure response times remain acceptable.

Question 5

A test case describes a specific _____, the steps to reproduce it, the _____ result, and the actual result. A bug report documents a defect including steps to _____, expected vs. actual behavior, and _____ level to help developers prioritize fixes.

Question 6

_____ testing runs automatically every time code is committed and checks that new changes do not break existing features. _____ testing involves real users testing the software in their own environment before release to validate it meets their needs. These are also called alpha and _____ testing.

Question 7

Security testing attempts to find _____ in the software before attackers do. _____ testing simulates attacks to find exploitable vulnerabilities. _____ analysis reviews source code for security flaws without executing it. A recently published list of the top 10 web application vulnerabilities is maintained by _____.

Question 8 -- Real World Application

REAL WORLD: A development team releases a new feature in their mobile banking app without running regression tests because they assumed the new code only affected the new feature. After release, customers report they can no longer log in because authentication broke. What type of testing should have caught this and describe how it works.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-4-3-testing-and-quality-assurance/tp-4-3-testing-and-quality-assurance-answer-key.pdf
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